Problem Set 15

Problem 12.6.1: $\psi_E = Ae^{-r/a_0}$.

(1) No (θ,ϕ)-dependence implies $\psi_E \propto Y_0^0$, so we must have $\ell = 0$ and $m = 0$.

(2) Therefore $\psi_E = R_{E,\ell=0} = \frac{1}{r}U_{E,0}$ which satisfies eqn. (12.6.5) with $\ell = 0$:

$$(r\psi_E)'' + \frac{2\mu}{\hbar^2}[E - V(r)](r\psi_E) = 0. \tag{1}$$

As $r \to \infty$, $V(r) \to 0$, which implies in this limit $(r\psi_E)' = A(1 - \frac{r}{a_0})e^{-r/a_0} \approx -(A/a_0)re^{-r/a_0}$, so $(r\psi_E)'' \approx (A/a_0^2)re^{-r/a_0}$. Therefore, eqn. (1) reads in this limit

$$\frac{A}{a_0^2}re^{-r/a_0} = -\frac{2\mu E}{\hbar^2}Are^{-r/a_0},$$

from which it follows that

$$E = -\frac{\hbar^2}{2\mu a_0^2}.$$

(3) Now plug E into (1) and use $(r\psi_E)'' = A(-\frac{2}{a_0} + \frac{r}{a_0^2})e^{-r/a_0}$ to get

$$A\left(-\frac{2}{a_0} + \frac{r}{a_0^2}\right)e^{-r/a_0} + \frac{2\mu}{\hbar^2}\left(-\frac{\hbar^2}{2\mu a_0^2} - V(r)\right)Are^{-r/a_0} = 0,$$

which gives

$$V(r) = -\frac{\hbar^2}{\mu a_0 r}.$$

Problem 12.6.4:

(1) $\delta^3(\mathbf{r}-\mathbf{r}')$ is defined by the property that $\int d^3r\delta^3(\mathbf{r}-\mathbf{r}')f(\mathbf{r}) = f(\mathbf{r}')$. So simply check:

$$\int r^2 dr\sin\theta d\theta d\phi \frac{1}{r^2\sin\theta}\delta(r-r')\delta(\theta-\theta')\delta(\phi-\phi')f(r,\theta,\phi)$$
$$= \int drd\theta d\phi\delta(r-r')\delta(\theta-\theta')\delta(\phi-\phi')f(r,\theta,\phi) = f(r',\theta',\phi').$$

(2) If $r \neq 0$ then $\nabla^2(\frac{1}{r}) = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}(\frac{1}{r}))+$(angular parts) $= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2(\frac{-1}{r^2})) = \frac{1}{r^2}\frac{\partial}{\partial r}(-1) = 0$. When $r = 0$ the above calculation breaks down since terms are singular there. So consider an arbitrary continuous function $f(\mathbf{r})$ and the integral $\int d^3x\nabla^2(\frac{1}{r})f(\mathbf{r}) = \lim_{\epsilon\to 0}\int_0^\epsilon r^2dr\int d\Omega\nabla^2(\frac{1}{r})f(\mathbf{r})$, since $\nabla^2(\frac{1}{r}) = 0$ for $r > 0$. Then, integrating by parts we get

$$\int d^3x\nabla^2\left(\frac{1}{r}\right)f(\mathbf{r}) = \lim_{\epsilon\to 0}\int_0^\epsilon r^2dr\int d\Omega\left\{\vec{\nabla}\cdot\left(f(\mathbf{r})\vec{\nabla}\left(\frac{1}{r}\right)\right) - \left(\vec{\nabla}f(\mathbf{r})\right)\cdot\vec{\nabla}\left(\frac{1}{r}\right)\right\}.$$

Now recall that in spherical coordinates $\vec{\nabla} = \widehat{r}\frac{\partial}{\partial r} + \widehat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta} + \widehat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}$, implying that $\vec{\nabla}(1/r) = \widehat{r}\frac{\partial}{\partial r}(1/r) = -\widehat{r}/r^2$, so

$$\int d^3x \nabla^2 \left(\frac{1}{r}\right) f = \lim_{\epsilon \to 0} \int_0^\epsilon r^2 dr \int d\Omega \left\{ \vec{\nabla}\cdot \left(-\widehat{r}\frac{f}{r^2}\right) + \left(\widehat{r}\cdot\vec{\nabla} f\right)\frac{1}{r^2} \right\}.$$

The second term on the right side vanishes, because as $\epsilon \to 0$, $\widehat{r}\cdot\vec{\nabla} f = (\partial f/\partial r)|_{r=0} =$ const., so:

$$\lim_{\epsilon\to 0}\int_0^\epsilon r^2 dr \int d\Omega \left(\widehat{r}\cdot\vec{\nabla} f\right)\frac{1}{r^2} = \text{const.}\cdot \lim_{\epsilon\to 0}\int_0^\epsilon r^2 dr \int d\Omega \frac{1}{r^2} = \text{const.}\cdot \lim_{\epsilon\to 0} 4\pi\epsilon = 0.$$

Therefore,

$$\begin{aligned}
\int d^3x \nabla^2 \left(\frac{1}{r}\right) f &= \lim_{\epsilon\to 0}\int_0^\epsilon r^2 dr \int d\Omega \vec{\nabla}\cdot\left(-\widehat{r}\frac{f}{r^2}\right) = \lim_{\epsilon\to 0}\int d\Omega r^2 \widehat{r}\cdot\left(-\widehat{r}\frac{f}{r^2}\right)\Big|_{r=\epsilon} \\
&= -\lim_{\epsilon\to 0}\int d\Omega\, f|_{r=\epsilon} = -f(0)\lim_{\epsilon\to 0}\left(\int d\Omega\right) = -f(0)\lim_{\epsilon\to 0} 4\pi \\
&= -4\pi f(0).
\end{aligned}$$

In the second step I used the divergence theorem which states $\int_R d^3x \vec{\nabla}\cdot\vec{g} = \int_{\partial R} d^2a\, \widehat{n}\cdot\vec{g}$ where R is any region, ∂R is its boundary, $\widehat{n}$ is the normal unit vector to ∂R (pointing out of R) and d^2a is the surface area element. In our case $R = \{r < \epsilon\}$, $d^2a = d\Omega$, $\widehat{n} = \widehat{r}$, and $\vec{g} = -\widehat{r}(f/r^2)$. Thus we have shown that $\nabla^2(1/r) = 0$ for $r \neq 0$ and for any $f(\vec{r})$ that $\int d^3x \nabla^2(1/r) f = -4\pi f(0)$. This is the *definition* of the delta function, so

$$\nabla^2\left(\frac{1}{r}\right) = -4\pi\delta^3(\mathbf{r}).$$

Problem 12.6.9: Since $\ell = 0$, $\psi = R(r)Y_0^0(\theta,\phi) = R(r)$. So the radial equation becomes, with $\psi(r) = (1/r)U(r)$,

$$\begin{aligned}
\left(\frac{d^2}{dr^2} + k^2\right) U_{\text{in}} = 0 \qquad & r \le r_0, \qquad k \equiv \sqrt{\frac{2\mu(E+V_0)}{\hbar^2}}, \\
\left(\frac{d^2}{dr^2} - \kappa^2\right) U_{\text{out}} = 0 \qquad & r \ge r_0, \qquad \kappa \equiv \sqrt{\frac{-2\mu E}{\hbar^2}},
\end{aligned}$$

where k and κ are defined to be the positive root. The solutions of these equations are

$$\begin{aligned}
U_{\text{in}} &= A\sin kr + \widetilde{A}\cos kr, \\
U_{\text{out}} &= Be^{-\kappa r} + \widetilde{B}e^{+\kappa r}.
\end{aligned}$$

The boundary conditions are that $U_{\text{in}} \to 0$ at $r = 0$, implying $\widetilde{A} = 0$, and $U_{\text{out}} \to 0$ at $r = \infty$, implying that $\widetilde{B} = 0$. The matching conditions at $r = r_0$ are the continuity of ψ and its first derivative:

$$\psi_{\text{in}} = \psi_{\text{out}} \qquad \Rightarrow \qquad U_{\text{in}} = U_{\text{out}}|_{r=r_0}, \tag{2}$$

$$\psi'_{\text{in}} = \psi'_{\text{out}} \qquad \Rightarrow \qquad (U_{\text{in}}/r)' = (U_{\text{out}}/r)'|_{r=r_0}. \tag{3}$$

Eqn. (2) implies $A \sin kr_0 = Be^{-\kappa r_0}$, or,

$$B = A \sin kr_0 \, e^{\kappa r_0}. \tag{4}$$

Eqn. (3) implies $\frac{d}{dr}[(A/r)\sin kr - (B/r)e^{-\kappa r}]_{r=r_0} = 0$, which gives

$$-\frac{A}{r_0^2}\sin kr_0 + \frac{Ak}{r_0}\cos kr_0 + \frac{B}{r_0^2}e^{-\kappa r_0} + \frac{B\kappa}{r_0}e^{-\kappa r_0} = 0.$$

Plugging (4) into this gives $(Ak/r_0)\cos kr_0 + (A\kappa/r_0)\sin kr_0 = 0$, or

$$-\tan kr_0 = \frac{k}{\kappa}, \tag{5}$$

which is what we wanted to show.

If $V_0 < \pi^2\hbar^2/(8\mu r_0^2)$ and $-V_0 < E < 0$ (for a bound state), then $k^2 = \frac{2\mu}{\hbar^2}(E + V_0) < \frac{2\mu}{\hbar^2}\pi^2\hbar^2/(8\mu r_0^2) = \pi^2/(4r_0^2)$. Thus $0 < kr_0 < (\pi/2)$, which implies that $-\tan kr_0 < 0$. On the other hand, by definition, $k/\kappa > 0$. So there is no solution to eqn. (5).